

Antoine Lucas

Biostatistician & Data Scientist

Pharma · Cosmetics · Biotech · Paris, France

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Biostatistician and Data Scientist with experience across pharma, cosmetics, and biotech. Builds scientific software, statistical workflows, simulation tools, and applied ML in R and Python, with a strong focus on readability, reproducibility, and stakeholder-facing delivery.

Selected Experience

Data Scientist & ML Engineer

Sep 2025 – present

Chanel Parfums Beauté R&D · via Astek / IT&M Stats

- Delivered computer-vision, NLP, and R Shiny tools for formulation R&D — texture and colour analysis, olfactory descriptor mining.
- Modelled sensory and physico-chemical properties with PLS, Gaussian processes, Bayesian optimisation, and DoE, cutting formulation experiments by 30–40%.
- Managed delivery and platform lifecycle across Azure ML, Databricks, Posit Connect, Docker, GitHub Actions, and a 20+ app catalog.

Data Scientist — Internal & Consulting

May – Sep 2025

Astek / IT&M Stats

- Built reusable R packages and Quarto reporting templates, reducing analyst turnaround by 25% across consulting engagements.
- Led internal training on tidymodels, causal inference, and reproducible research for 8 statistical consultants.
- Scoped pre-sales ML opportunities and fine-tuned a Phi-3 model for claim verification from data preparation through evaluation.

Data Scientist — Multi-omics

Dec 2024 – May 2025

Abolis · Microbiome Studio · via Astek / IT&M Stats

- Worked autonomously in an 8-person startup building a SaaS product for multi-omics analysis.
- Delivered end-to-end R and Python pipelines for genomics, transcriptomics, metabolomics, and tabular datasets.
- Applied PERMANOVA, sparse PLS-DA, and network analysis; integrated outputs into the production platform via REST APIs.

Data Scientist

Jul 2023 – Nov 2024

Sanofi R&D · CMC Data Science · via Astek / IT&M Stats

- Developed R and Python applications for multiple user profiles across CMC data-science workflows.
- Built simulation tools for global resource allocation and CMC plant-capacity planning across production sites.
- Supported BioProcess Drug Product activities with predictive tools, including particle-load modelling for monoclonal antibodies.

Biostatistician

Nov 2021 – Jul 2023

L'Oréal R&I · Scientific Computing · via Astek / IT&M Stats

- Supported statistical analyses of exposome and clinical datasets: biophysical, omics, and microbiome data.
- Characterised clinical phenotypes with multi-block, integrative, multivariate, and variable-selection methods; developed Bayesian-network visualisations.
- Defined methodologies for POC clinical studies; wrote SAPs, supported blind reviews, programmed analyses in R, and reported results.

Earlier Experience

Data Scientist Intern — L'Oréal R&I · Augmented Beauty

Feb – Aug 2021

Modelled hair and face customer data for personalised recommendations through data management, statistical analyses, machine learning, and communication to R&I teams.

Biostatistics Project Officer Intern — Da Volterra · Metagenomics Team

Feb – Aug 2020

Performed data management and non-parametric analyses for gut-microbiome clinical studies; prepared and presented results to biotechnology research teams.

Data Scientist Intern — Flinders University · Digital Health Research Center

Sep 2019 – Jan 2020

Supported digital-health and wearables research in R: questionnaire design, workshop preparation, statistical evaluation, and presentation of results.

Education

Diplôme d'Ingénieur — Statistiques appliquées aux Sciences de la Vie (2021)

Institut Agro - Agrocampus Ouest, Rennes

Master — Mathématiques Appliquées · Statistiques (2021)

INSA Rennes · Agrocampus Ouest · Université Rennes 2 · ENSAI · ENSAE (joint programme)

Technical Skills

Programming	R · Python · SQL · Bash
ML & Stats	scikit-learn · tidymodels · XGBoost · Bayesian optimisation · DoE · PLS · PERMANOVA · survival analysis
Apps & Viz	R Shiny (golem) · ggplot2 · Plotly · Quarto · Power BI
CV / NLP	PyTorch · OpenCV · Hugging Face · sentence-transformers
Infrastructure	Docker · GitHub Actions · Posit Connect · Azure ML · Databricks · renv · uv
Domains	GxP · Clinical biostatistics · Multi-omics · Cosmetic R&D

Additional Information

Certifications

GitHub Foundations — GitHub (2024)

Scrum Master — Agilbee (2023)

Languages

French (native)

English (fluent, professional)